

What is Computer?

Computer:- A computer is an electronic device that accepts data as input, processes it according to a set of instructions, produces meaningful information as output and stores the results for future use.

A computer performs four basic operations:

Input - Entering data and instructions into the computer.

Processing - Manipulating the data according to given instructions.

Output - Producing the processed result in a meaningful form.

Storage - Storing data, instructions and results for future use.

Basic Working of a Computer

Input → Processing → Output → Storage

Characteristics of computer.

The important characteristics of a computer are:

1. Speed

A computer can perform a large number of calculations and operations in a very short period of time. It can process data much faster than humans.

2. Accuracy

A computer produces highly accurate results when the data and instructions supplied to it are correct. Errors generally occur due to incorrect input or instructions.

3. Diligence

A computer does not become tired or bored. It can perform the same task repeatedly with the same speed and accuracy.

4. Storage Capacity

A computer can store a large amount of data, programs and information. The stored information can be retrieved whenever required.

5. Versatility

A computer can perform different types of tasks. It can be used for calculations, preparing documents, communication, education, business, entertainment and many other purposes.

6. Automation

Once a program and the required instructions are given, a computer can perform operations automatically without continuous human intervention.

7. Reliability

A computer can perform repeated operations consistently and provide reliable results, provided that the instructions and input data are correct.

8. Multitasking

A computer can handle multiple tasks or programs at the same time. For example, a user can listen to audio while working on a document.

9. Large Processing Capacity

A computer can process a large volume of data efficiently and in a relatively short time.

Applications of computer.

Computers are used in almost every field of life. Some important applications of computers are as follows:

1. Education

Computers are used for online classes, preparing notes, presentations, examinations, research and maintaining student records.

2. Business

Businesses use computers for accounting, billing, maintaining records, preparing reports, communication and managing business operations.

3. Banking

Computers are used to maintain customer accounts, process transactions, prepare statements and provide services such as ATM and online banking.

4. Healthcare

Computers are used for maintaining patient records, medical research, diagnosis, hospital management and various laboratory activities.

5. Communication

Computers facilitate communication through email, video conferencing, messaging and other online communication services.

6. Science and Research

Scientists use computers for calculations, data analysis, simulations, experiments and storing research data.

7. Government Offices

Computers are used for maintaining official records, preparing documents, managing databases and providing various public services.

8. Entertainment

Computers are used for watching movies, playing games, listening to music, creating digital content and other forms of entertainment.

9. Transportation

Computers are used in railway and airline reservation systems, traffic management, navigation and vehicle control systems.

10. Industry

Industries use computers for designing, production planning, automation, quality control and maintaining records.

Conclusion

Thus, computers have become an essential part of modern life and are widely used in education, business, banking, healthcare, communication, science and many other fields.

Digital and Analog Computer.

Digital computer: - A digital computer is a computer that processes data in the form of discrete values, generally represented by binary digits 0 and 1. It performs arithmetic and logical operations on digital data.

Examples: Desktop, laptop, digital calculator and smartphone.

Features of Digital Computer

- Processes discrete data.
- Uses binary numbers (0 and 1).
- Gives accurate and precise results.
- Suitable for arithmetic and logical operations.
- Widely used in homes, offices, schools and businesses.

Analog Computer: - An analog computer is a computer that processes data in the form of continuous physical quantities such as temperature, pressure, speed and voltage.

It is mainly used for measuring and monitoring continuously changing physical conditions.

Examples: Analog speedometer, analog thermometer and certain industrial control systems.

Features of Analog Computer

- Processes continuous data.
- Measures physical quantities.
- Results are represented in a continuous form.
- Useful for real-time measurement and monitoring.
- Used in scientific and industrial applications.

Generations of Computer.

The development of computers can be divided into different generations on the basis of the major technology used in their construction and operation. There are five generations of computers.

1. First Generation (1940-1956)

The first generation of computers used vacuum tubes for processing and magnetic drums for memory.

Main Features

Used vacuum tubes.

Very large in size.

Consumed a lot of electricity.

Produced considerable heat.

Less reliable compared with later generations.

Programming was mainly done in machine language.

Examples

ENIAC

UNIVAC I

EDVAC

2. Second Generation (1956-1963)

The second generation replaced vacuum tubes with transistors.

Main Features

Used transistors.

Smaller than first-generation computers.

Faster and more reliable.

Consumed less electricity.

Produced less heat.

Assembly language and some high-level languages were used.

Examples

IBM 1401

CDC 1604

3. Third Generation (1964-1971)

The third generation used Integrated Circuits (ICs) instead of individual transistors.

An IC contains many electronic components on a small chip.

Main Features

Used Integrated Circuits.

Smaller and faster.

More reliable.

Less expensive compared with earlier generations.

Consumed less power.

High-level programming languages became more common.

Examples

IBM System/360

PDP-8

4. Fourth Generation (1971-Present)

The fourth generation introduced microprocessors. A microprocessor placed the processing functions of a computer on a single chip.

Main Features

Used microprocessors.

Computers became much smaller.

Very high processing speed.

More reliable and affordable.

Personal computers became common.

Large memory and storage became available.

Examples

Personal Computers

Desktop computers

Laptops

5. Fifth Generation (Present and Beyond)

The fifth generation is associated with Artificial Intelligence (AI) and advanced computing technologies.

The main aim is to develop computers capable of performing tasks that require aspects of human intelligence, such as learning, reasoning and understanding natural language.

Main Features

Based on Artificial Intelligence.

Uses advanced processors and computing technologies.

Supports natural language processing.

Focuses on intelligent systems.

Used in robotics, expert systems and AI-based applications.

Examples

AI systems

Intelligent robots

Voice based assistant

Types of computer:-

1. **Super computer:-** A super computer is a very powerful computer designed to perform extremely complex calculations and process huge amount of data at very high speed.

Uses

Supercomputers are mainly used where very large and complex calculations are required

Weather forecasting

Climate research

Space research

Scientific simulations

PARAM SERIES OF SUPERCOMPUTER IN INDIA

2. **Mainframe computer:-** A Mainframe computer is a powerful computer designed to process large volume of data and support many users and applications at the same time.

Uses

Banking

Railway station

Government offices

Insurance companies

3. **Mini computer:-** A mini computer is a computer that is smaller and generally less powerful than a mainframe computer but more capable than a typical personal computer in traditional classifications.

Uses

Scientific laboratories

Small and medium organisations

Industrial control

Business applications

Department level data processing

Memory Hierarchy:-

Meaning of Memory Hierarchy

A computer uses different types of memory to store data and instructions. These memories differ in their speed, storage capacity and cost. Therefore, computer memory is arranged in different levels according to their performance and accessibility. This arrangement is called memory hierarchy.

The main purpose of memory hierarchy is to provide the computer with fast access to frequently required data while also providing large storage capacity.

The main levels of memory hierarchy are:

Registers → Cache Memory → Main Memory → Secondary Storage

1. Registers

Registers are the smallest and fastest storage locations in a computer. They are located inside the CPU.

Registers temporarily hold the data, instructions and intermediate results that are currently being processed by the CPU.

For example, when the CPU performs a calculation, the values required for the calculation may be temporarily stored in registers.

Main features:

They are extremely fast.

They have very small storage capacity.

They are directly used by the CPU.

They store data and instructions currently being processed.

Example: Accumulator, Program Counter and Instruction Register.

2. Cache Memory

Cache memory is a high-speed memory that stores frequently used data and instructions.

It is located very close to the CPU or integrated with the processor. Since the CPU can access cache faster than main memory, it helps to reduce the time required to access frequently used information.

For example, if the CPU repeatedly needs the same data, keeping that data in cache can allow it to be accessed more quickly.

Main features:

Very fast memory.

Faster than main memory.

Smaller capacity than RAM.

Stores frequently used data and instructions.

Improves the overall performance of the computer.

Examples: L1, L2 and L3 cache.

3. Main Memory

Main memory is the memory that the computer uses to store data and instructions that are currently required for processing.

It mainly consists of RAM and ROM.

RAM — Random Access Memory

RAM is a volatile memory, which means that its contents are normally lost when the power is switched off.

It stores currently running programs, data and instructions.

Example: When you open a browser or a document, the required data is loaded into RAM.

ROM — Read Only Memory

ROM is non-volatile memory. Its contents are retained when the power is switched off.

It stores important instructions required for starting or operating the computer.

Main features of Main Memory:

Faster than secondary storage.

Directly accessible by the CPU.

Larger than cache and registers.

Smaller than most secondary storage devices.

4. Secondary Storage

Secondary storage is used to store data and programs for a long period of time.

Unlike RAM, it is non-volatile, so data remains stored even when the computer is switched off.

Examples:

Hard Disk Drive (HDD)

Solid State Drive (SSD)

Pen Drive

Memory Card

Main features:

Large storage capacity.

Non-volatile.

Slower than main memory.

Used for permanent or long-term storage.

Generally cheaper per unit of storage than faster memory.

Storage units:- Storage units are used to measure the amount of data that a computer can store. The basic unit of digital information is a bit, which can have a value of 0 or 1. Eight bits make one byte. Larger units include Kilobyte, Megabyte, Gigabyte, Terabyte and Petabyte.

Units	Description
Bit	Consists of 1 or 0
Byte	Consists of 8 bits
Kilobyte	Consists of 1024 bytes
Megabyte	Consists of 1024 kilobytes
Gigabyte	Consists of 1024 megabytes
Terabyte	Consists of 1024 gigabytes
Petabyte	Consists of 1024 terabytes
Exabyte	Consists of 1024 petabytes
Zettabyte	Consists of 1024 exabytes
Yottabyte	Consists of 1024 zettabytes

Input devices:- Input devices are hardware devices used to enter data and instructions into a computer for processing. They convert the user's actions or information into a form that can be understood and processed by the computer.

1. Keyboard

A keyboard is one of the most common input devices. It contains keys used to enter letters, numbers, symbols and commands into a computer.

Uses:

Typing text

Entering numbers

Giving commands

Using shortcut keys

Example: QWERTY keyboard.

2. Mouse

A mouse is a pointing device used to control the movement of the pointer on the computer screen.

Uses:

Selecting items

Opening files

Clicking buttons

Dragging objects

Scrolling pages

3. Pointing and Drawing Devices

These devices are used to point, select or draw objects on the computer screen.

Touchscreen

A touchscreen allows the user to interact directly with the screen by touching it.

Examples: Smartphones and tablets.

Light Pen

A light pen is a pointing device used to select or draw objects directly on a suitable computer display.

Graphics Tablet

A graphics tablet allows users to draw images or designs digitally using a stylus.

4. Data Scanning Devices

Data scanning devices capture information from physical documents, images or other sources and convert it into a form that can be processed by a computer.

Scanner

A scanner captures text or images from paper documents and converts them into digital form.

Barcode Reader

A barcode reader reads barcodes printed on products and sends the information to a computer.

Example: Supermarket billing counters.

OCR

OCR stands for Optical Character Recognition. It recognizes printed or handwritten characters from an image and converts them into editable digital text.

OMR

OMR stands for Optical Mark Recognition. It detects marked areas on specially designed forms.

Example: Objective-type answer sheets.

MICR

MICR stands for Magnetic Ink Character Recognition. It is used to read characters printed with special magnetic ink.

Example: Bank cheques.

5. Microphone

A microphone is an input device that captures sound and sends it to a computer for processing.

Uses:

Voice recording

Online meetings

Voice commands

Speech recognition

6. Camera/Webcam

A digital camera or webcam captures images and video and provides them as input to a computer.

Uses:

Video calls

Online classes

Photography

Video recording

Facial recognition systems

Conclusion

Input devices are an essential part of a computer system because they allow users to enter data and instructions into the computer. Different input devices are used for different types of information such as text, images, sound and marks.

Output Devices:- Output devices are hardware devices used to display, present or produce the results obtained after processing data by a computer. They convert the processed information into a form that can be understood by the user.

1. Monitor

A monitor is the most common output device. It displays text, images, videos and other information on the screen.

Uses:

Displaying documents

Watching videos

Viewing images

Displaying computer programs and results

Examples: LCD, LED and OLED monitors.

2. Printer

A printer is an output device that produces a hard copy of text, images or other information on paper.

Types of Printers

a) Impact Printers

Impact printers produce output by physically striking an inked ribbon against paper.

Example: Dot Matrix Printer.

b) Non-Impact Printers

Non-impact printers produce output without physically striking the paper.

Examples: Inkjet Printer and Laser Printer.

Uses:

Printing documents

Printing photographs

Printing reports and assignments

3. Speaker

Speakers are output devices that produce sound from a computer.

Uses:

Listening to music

Watching videos

Online classes

Video conferencing

Hearing alerts and notifications

4. Projector

A projector displays the computer's output on a large screen or wall.

Uses:

Classroom teaching

Presentations

Meetings

Seminars

Public demonstrations

5. Plotter

A plotter is an output device used to produce high-quality drawings, designs and large graphical outputs on paper.

Uses:

Engineering drawings

Architectural designs

Maps

Large technical drawings

6. Headphones

Headphones are output devices that allow a user to hear sound privately from a computer.

Uses:

Listening to audio

Online classes

Video calls

Watching videos

Difference Between Soft Copy and Hard Copy

Soft Copy: Information displayed electronically on a screen.

Example: A document displayed on a monitor.

Hard Copy: Information printed on physical paper.

Example: A printed document from a printer.

Output devices enable the user to receive the results processed by a computer in different forms such as visual, printed and audio form. They are therefore an important part of a computer system.

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